

International Trade: Theory and Policy

國際貿易: 理論與政策

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What is International Trade?

Countries are **interdependent**, and how they interact economically profoundly affects prosperity, jobs, prices, innovation, inequality, and international relations. Ignoring it is like trying to understand a modern economy while pretending the rest of the world doesn't exist.

Gains from Trade (The Fundamental Reason)

- Countries **are not equally** good at producing everything. Through **comparative advantage** (David Ricardo's classic insight), countries can **specialize** in what they produce relatively more **efficiently** and trade for the rest.
- Result: More total output, lower prices for consumers, and higher living standards for everyone involved — even if one country is absolutely better at producing everything.

Access to Goods, Resources, and Ideas

- No country has **all** the natural resources, technology, or climate needed for everything its citizens want (e.g., Japan imports oil and food; Saudi Arabia imports technology and cars).
- Trade gives consumers **variety and lower costs** (smartphone, clothes, coffee, medicine, etc., are cheaper and better because of global supply chains).

Economic Growth and Development

- Trade expands market size → encourages innovation, economies of scale, and competition.
- Historically, countries that opened up to trade (East Asia in the late 20th century, China after 1978, post-WWII Western Europe) experienced rapid growth and poverty reduction.
- Developing countries often use trade as a ladder out of poverty.

Global Challenges

- Climate change (carbon leakage, border carbon adjustments)
- Supply chain resilience (COVID-19 exposed vulnerabilities)
- National security (critical minerals, semiconductors, rare earths)
- Global poverty and development (trade vs. aid debate)

- **For individuals:** Better understanding of job markets, investment opportunities, and personal finance in a globalized world.
- **For businesses:** Deciding where to source, produce, or sell.
- **For policymakers:** Designing better economic strategies.
- **For citizens:** Making informed voting decisions instead of falling for simplistic slogans ("trade is always good" or "trade is always bad").

International Trade and Domestic Trade

Domestic trade is like trading between Nantou and Tainan – same currency, same federal laws, easy movement of people. **International trade** is like trading between the US and Taiwan – different rules, different money, borders, politics, and security concerns ◦

Because of these frictions, international trade is more complex and politically controversial than domestic trade. This is why we have entire fields of study (International Economics) and institutions (WTO, IMF, World Bank) dedicated to managing it.

Importance of International Trade

International trade is a cornerstone of the modern global economy. It drives **growth**, **efficiency**, **innovation**, and **human development** on a massive scale.

Economic Growth and Efficiency

- Trade allows countries to **specialize** based on **comparative advantage**, leading to higher overall productivity and output.
- It **expands market size**, enabling **economies of scale** and greater competition, which spurs innovation and efficiency.
- **Global scale:** World trade (goods + services) typically represents **55–65% of global GDP**. In recent years, trade volume has often grown faster than GDP, amplifying economic expansion.

Lower Prices and Greater Consumer Choice

- Imports increase **competition** and supply, reducing prices for everyday goods (electronics, clothing, food, medicine).
- Consumers gain access to a **variety of products** and **higher-quality goods** that might not be available or affordable domestically.

Job Creation and Income Generation

- **Export** sectors create high-quality jobs (manufacturing, agriculture, services, tech).
- **Global supply chains** support millions of jobs worldwide — from farmers in Brazil to factory workers in Asia to engineers in the US/Europe.

Innovation, Technology Transfer, and Knowledge Spillovers

- Exposure to international markets encourages firms to **adopt new technologies** and best practices.
- **Foreign direct investment (FDI)** and trade in high-tech goods/services (e.g., AI-related trade) accelerate innovation.

Real-World Evidence (Recent Context)

- In 2025, global trade showed resilience despite tensions, with strong growth in **AI-related** and advanced manufacturing goods.
- Trade continues to be a key engine even amid geopolitical shifts, though it faces headwinds from tariffs and uncertainty.

Not Automatic Benefits

While the overall gains are large, trade also creates **winners** and **losers** within countries (e.g., some industries decline). This is why **complementary policies** — education, worker retraining, social safety nets — are essential to maximize benefits and manage adjustment costs.

Thoery of International Trade

International trade theories have evolved over time to better explain real-world patterns. Here's a structured overview of **Traditional (Classical)**, **New Trade Theory**, and **New New Trade Theory** ("New-New") ◦

Traditional Trade Theories (18th–19th Century)

These are **country-level** theories focused on **why countries specialize and trade** based on differences between them.

- **Absolute Advantage (Adam Smith, 1776)** Countries should specialize in and export goods they can produce **more efficiently** (using fewer resources) than other countries. **Why trade?** Mutual benefit through higher total output. Example: One country is better at wine, another at cloth → both gain by specializing and trading.
- **Comparative Advantage (David Ricardo, 1817)** Even if one country is better at producing *everything* (absolute advantage), trade still benefits both if they specialize in goods with the **lowest opportunity cost**.
- **Key insight:** Trade is driven by **relative efficiency** differences, not absolute ones. This is the foundation of modern trade theory and explains why trade occurs even between unequal partners.

- **Heckscher-Ohlin (H-O) Model (20th Century extension)** Countries export goods that intensively use their **abundant factors** of production (e.g., labor-abundant countries export labor-intensive goods; capital-abundant export capital-intensive ones).
- **Limitations of Traditional Theories:** They assume **perfect competition**, **homogeneous products**, **constant returns to scale**, and **no trade between similar countries**. They struggle to explain **intra-industry trade** (e.g., Germany and the US both exporting and importing cars).

New Trade Theory (Late 1970s–1980s)

Developed mainly by **Paul Krugman** (Nobel Prize 2008). It addresses the shortcomings of traditional theories by incorporating **economies of scale** and **imperfect competition**.

Core Ideas:

- **Increasing Returns to Scale** (Economies of Scale): Larger production lowers average costs. Countries specialize to achieve scale, even without comparative advantage differences.
- **Monopolistic Competition + Product Differentiation**: Firms produce **differentiated** (branded) varieties. Consumers **love variety** → countries trade similar goods (intra-industry trade).
- **Home Market Effect**: **Larger** domestic markets attract more firms, leading to exports.

Why trade according to NTT? Trade allows firms to exploit economies of scale and meet consumer demand for variety. Much of modern trade (e.g., cars, electronics, aircraft between rich countries) is explained here, not just by factor endowments.

This theory also opened the door for strategic trade policy debates (governments potentially helping "national champions").

New New Trade Theory (2000s–present)

Also called **Heterogeneous Firms Trade Theory**, pioneered by **Marc Melitz** (2003) and others. It shifts focus from countries or industries to **individual firms**.

Core Ideas:

- Firms are **heterogeneous (different productivity levels)**, not identical.
- Only the **most productive firms** can afford the fixed costs of exporting (e.g., market research, distribution, compliance).
- Trade liberalization → **self-selection**: High-productivity firms export and grow; low-productivity firms shrink or exit → raises average industry productivity.
- Explains **Export vs. FDI** decisions: Very productive firms invest abroad; moderately productive ones export.

Why trade?

International trade **reallocates resources** toward more efficient firms, drives productivity gains, innovation, and industry dynamics. It explains why only a small percentage of firms export, even in open economies.

Theory	Level of Analysis	Main Drivers of Trade	Key Explanation for "Why Trade?"	Real-World Strength
Traditional	Country	Differences in productivity or factors	Comparative advantage & specialization	North-South trade, resource-based
New (Krugman)	Industry/Firm	Economies of scale + variety	Scale + consumer love for differentiated products	Intra-industry trade between similar countries
New New (Melitz)	Individual Firm	Firm productivity differences	Selection of productive firms + reallocation	Firm-level data, export behavior

These theories are complementary, not contradictory. Traditional theories explain broad patterns between dissimilar countries, while New and New New theories better explain trade among similar advanced economies and firm-level realities in a globalized world.

Trade Policy

Trade policy refers to **government actions** that influence the flow of goods and services across borders. The **two main categories** of trade barriers (protectionist tools) are tariffs and non-tariff barriers

Tariffs

Tariffs are **taxes (duties) imposed on imported goods** at the border. They make foreign products more expensive, aiming to protect domestic industries, generate government revenue, or address unfair practices (e.g., dumping).

Types of Tariffs

- **Specific Tariff:** Fixed fee per unit (e.g., \$5 per barrel of oil).
- **Ad Valorem Tariff:** Percentage of the good's value (e.g., 25% on imported cars).
- **Compound Tariff:** Combination of both.
- **Tariff-Rate Quota (TRQ):** Low tariff up to a quota limit, then higher tariff.

Economic Effects of Tariffs

- **Raises prices** for consumers and businesses that rely on imports.
- **Protects domestic producers** (higher prices → more domestic sales and jobs in protected sectors).
- Revenue for the government.
- **Deadweight loss**: Overall economic **inefficiency** (higher prices reduce consumption; resources are misallocated).
- Can lead to **retaliation** (trade wars), hurting exporters.
- Incidence: Mostly borne by domestic consumers/importers (often near 100% pass-through).

Recent examples (2025–2026) show tariffs raising consumer prices (e.g., in goods like clothing, electronics, and durables) and generating significant revenue but also contributing to inflationary pressures and lower GDP growth in some models

Non-Tariff Barriers (NTBs or NTMs)

Non-tariff barriers are all other measures (**besides tariffs**) that restrict or distort trade. They are often more opaque, harder to measure, and potentially more restrictive than tariffs.

Common Types of NTBs

- **Quotas:** Direct limits on the **quantity** or value of imports.
- **Import Licensing:** Requirements for permits or approvals.
- **Voluntary Export Restraints (VERs):** Exporter agrees to limit shipments (often under pressure).
- **Technical Barriers to Trade (TBT):** Standards, testing, labeling, certification (e.g., safety or environmental requirements).
- **Sanitary and Phytosanitary (SPS) Measures:** Health and safety rules for food, animals, plants.
- **Subsidies:** Government support to domestic producers (makes imports relatively less competitive).
- **Local Content Requirements:** Must use a certain % of **domestic inputs**.
- **Administrative/Customs Procedures:** Delays, complex bureaucracy, or inspections.
- **Government Procurement Rules:** "Buy national" preferences.
- **Others:** Embargoes, anti-dumping duties, currency controls.

Economic Effects of NTBs

- Increase compliance costs and uncertainty for exporters.
- Can be disguised as legitimate regulations (health, safety, environment).
- Often favor domestic firms more subtly than tariffs.
- Higher impact on small/medium enterprises due to fixed compliance costs.
- Can reduce trade volumes more than equivalent tariffs in some cases.

Tariffs vs. Non-Tariff Barriers

Aspect	Tariffs	Non-Tariff Barriers
Nature	Transparent tax on imports	Often regulatory, less transparent
Visibility	Easy to measure and negotiate	Harder to quantify and challenge
Revenue	Generates government revenue	Usually does not (except some fees)
Flexibility	Can be adjusted quickly	More persistent and discretionary
WTO Discipline	Heavily regulated (bound rates)	Subject to rules (TBT, SPS agreements) but more loopholes
Modern Relevance	Still important (e.g., recent U.S. tariffs)	Increasingly dominant as tariffs fell post-WTO

Key Trend: After decades of tariff reductions through GATT/WTO, NTBs have become the primary tool (and concern) in trade policy.

Why Governments Use Them

Protectionism:

- Shield **infant industries** or declining sectors.
- **National security** (e.g., steel, semiconductors).
- **Political reasons** (protect jobs in key regions).
- Correct market failures (e.g., environmental standards, unfair subsidies abroad).

Downsides:

- Higher consumer prices and lower variety.
- Reduced efficiency and innovation due to less competition.
- Retaliation and trade wars.
- Harm to export sectors and global supply chains.
- Regressive impact (hits lower-income households harder).

Evidence

Overall Global Trade Picture (2025-2026)

Global trade reached a record **\$35 trillion in 2025**, growing approximately **7.5%** year-over-year unctad.org. World goods and services trade grew around **4.7%** in volume terms, significantly exceeding world GDP growth of 2.9%.

Manufacturing (Merchandise/Goods Trade)

Global Performance

- World merchandise trade volume grew 4.6% in 2025, well above earlier forecasts .
- The current U.S. dollar value of world merchandise trade (exports) was US\$26.26 trillion in 2025, up 7% compared to 2024 .
- Growth is projected to slow to **1.9% in 2026** under baseline conditions

Region/Country	2025 Export Growth	2025 Import Growth	Key Drivers
Asia	+9.5%	+6.0%	AI-related goods, semiconductors, electronics
China	+9.2%	—	AI-enabling products, manufacturing capacity
North America	+3.1%	+3.1%	Frontloading ahead of tariffs, gold, pharmaceuticals
Europe	-0.5%	+2.1%	Recovery from energy crisis, catch-up demand
Middle East	+12.9%	+10.4%	Energy exports
Africa	+10.3%	+8.7%	Commodity exports
South America	+3.2%	+10.4%	Agricultural and mineral exports

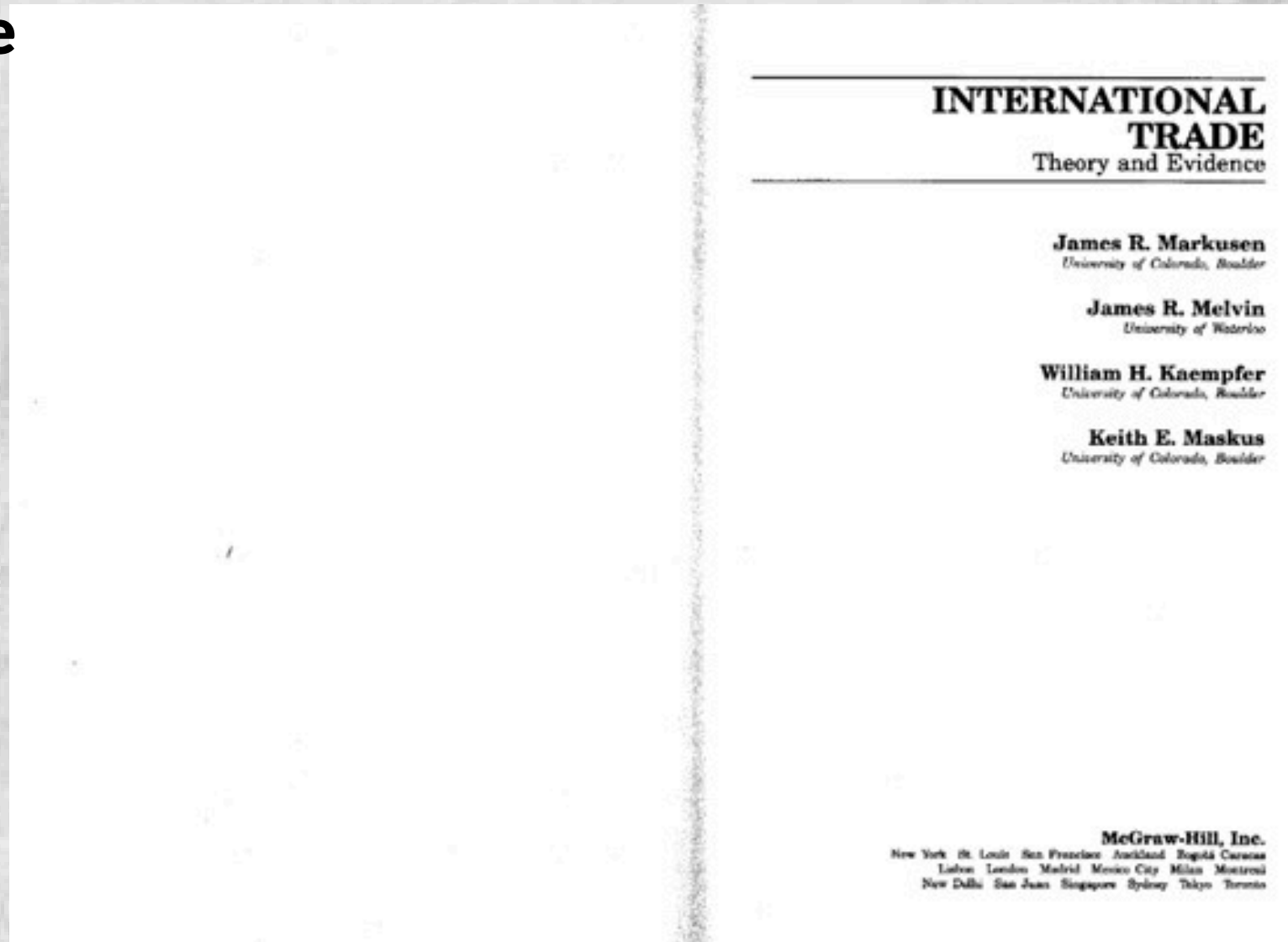
- While **agriculture** represents the physical foundation of global trade (food security, raw materials), **services** act as the invisible engine that makes modern agricultural trade possible (**logistics, digital tracking, trade finance**).
- In 2025, services trade grew at nearly double the rate of goods trade (~9% vs. ~4.6%), fundamentally transforming agriculture from a traditional commodity business into a highly optimized, data-driven global industry.

The following table contrasts the macroeconomic scale and trajectory of agricultural goods versus global services.

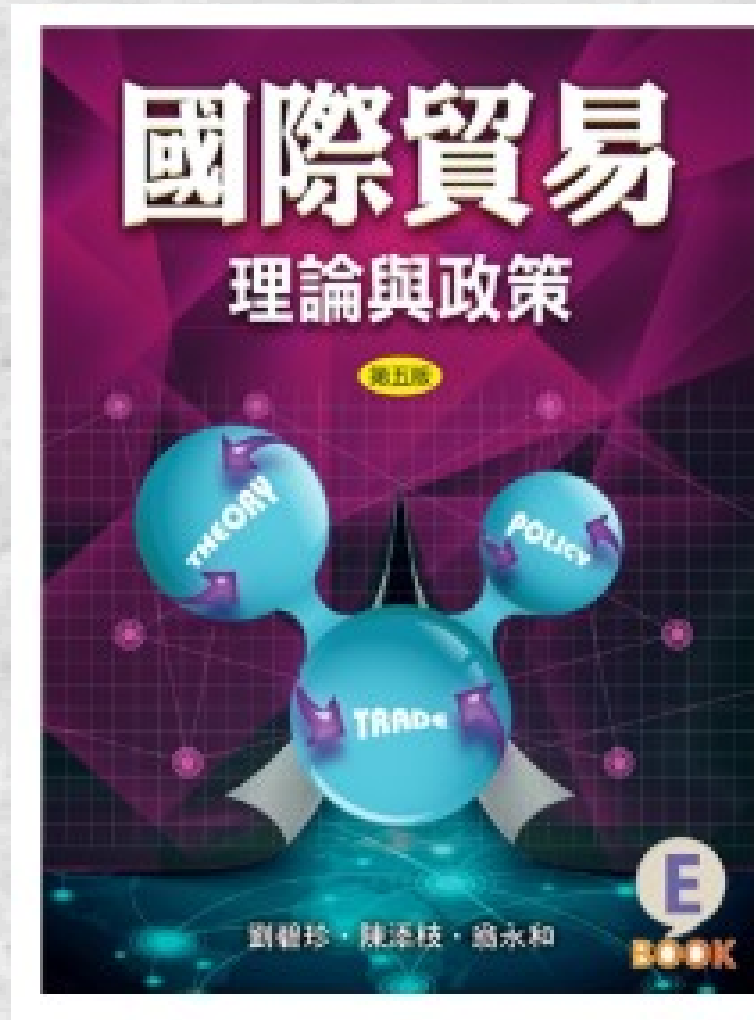
Metric	Agriculture (Merchandise/Goods)	Services Trade
Global Trade Value (2025)	Part of \$26.26 Trillion (Total Goods) \$9.56 Trillion (approx. 27% of total trade)	
2025 Volume/Value Growth	+4.6% (Goods overall)	~9.0% (Outpacing goods significantly)
2026 Growth Forecast	+1.9% (Slowing due to tariffs/geopolitics)	+4.8% (Moderating but remaining robust)
Primary Growth Drivers	Geopolitics, food security, climate/weather	Digitalization, AI, travel recovery, ICT
U.S. Trade Position (2025)	Exports: \$175.6B Imports: \$219.4B	Exports: \$1.209T Imports: \$870B

United States	Top Tier Exporter (\$175.6B). Major supplier to Asia and North America.	Global #1 Exporter (\$1.2T). Dominates digital, financial, and IP services.	Dual superpower. Massive services surplus offsets structural goods/ag deficits.
China	Top Tier Importer. Buys billions in U.S./global ag to ensure domestic food security.	Fastest Growing (+8.2% in Q4 2025). Driven by travel, transport, and ICT.	Transitioning to a service-driven model while securing physical food imports.
European Union	High-value, quality-focused agri-food exporter (e.g., dairy, wine, processed foods).	Mature Powerhouse (UK, Ireland, Germany). Strong in finance and business services.	Uses advanced services (legal, compliance) to enforce strict ESG standards on ag imports.
Brazil & Argentina	Commodity Giants. Top global exporters of soy, coffee, beef, and corn.	Growing, but heavily focused on supporting physical exports (basic logistics, transport).	Highly reliant on global transport and financial services to move bulk commodities.
India	Significant ag exporter (rice, spices) and growing processed food sector.	Surging Growth in both services exports and imports (IT, software, business processes).	Leveraging IT services dominance to modernize and optimize domestic agricultural supply chains.

Reference



Reference



Assessment Methods

- Midterm Exam 50% and Final Exam 50%
- **Closed-book** Multiple-Choice Questions (20 questions)

Format of Classes: Lecture

- Course Materials on Moodle and download free